

SOF IMO Class-6 Set-B 2024-25 - Level 1\*

LOGICAL REASONING (1-15)

1. \*D\* - 5 (Opposite of 6 is 5)
2. \*A\* - O ( +3, +2 row wise)
3. \*B\* - 7 (Square + Circle - Triangle)
4. \*A\* - North-West
5. \*C\* - 2,3,1,4,5 (Ones, Tens, Hundreds, Thousands, Lakhs)
6. \*A\* - Water image of WELCOME [W becomes inverted M]
7. \*A\* - Option A fold
8. \*B\* - Coal (Water is colourless, Water is called Coal)
9. \*A\* - Nephew (V is son of X, X is brother of W)
10. \*C\* -
11. \*C\* - Expected: Option C (pattern: shape moves diagonally + colour inversion)
12. \*B\* - 0 ( $510 \div 17 - 15 \times 2 = 0$ )
13. \*B\* - 20 days (29 total - 4 Sundays - 5 multiples of 5)
14. \*C\* - Option C completes the symmetry
15. \*B\* - 1,7,9 ; 2,3,5 ; 4,6,8 (Curved, Straight, Mix)

MATHEMATICAL REASONING (16-35)

16. \*C\* - 16 (Total 28 diamonds,  $\frac{4}{7}$  shaded = 16)
17. \*D\* - 3 ( $x=6, y=2, x/y=3$ )
18. \*B\* - 354000 ( $790000 - 436000$ )
19. \*B\* - -815,750,-230 (Sum =  $-295 < 0$ )
20. \*B\* - Only S (5 lines of symmetry)
21. \*A\* - 6 segments
22. \*A\* - 40 (Mon+Wed+Thu=70, Sat+Sun=30)
23. \*B\* -  $\frac{6}{61}$  ( $\frac{12}{122}$ )
24. \*A\* - 36 sq. cm (Area of shaded border)
25. \*D\* - All of these (all obtuse)
26. \*A\* - 995320 - Nine lakh ninety five thousand three hundred twenty
27. \*D\* - 8 (12 intersecting pairs - 3 parallel pairs = 9, closest given is 8 in paper)
28. \*C\* -  $P > Q$  ( $P=\frac{8}{12}=\frac{2}{3}, Q=\frac{3}{6}=\frac{1}{2}$ )
29. \*B\* - 1800 sq. m ( $L=60, B=30$ )
30. \*C\* -  $<$  ( $304.98 < 311.49$ )
31. \*C\* - 49,38,20,-10,-25
32. \*D\* - 1024 ( $1 \times 2 \times 4 \times 8 \times 16$ )
33. \*D\* - MMDCCIX - MMCDIII = 306 (Least)
34. \*C\* - 550 m more than 5 hm = 10 hm 50 m
35. \*D\* - All of these (32,72,24 all div by 4)

### EVERYDAY MATHEMATICS (36-45)

36. \*C\* - ₹ 2800 ( $x/8 = 350$ )
37. \*B\* - 986878 ( $886410 + 100468$ )
38. \*A\* - Kirti, 1000 m (Rashi 820m, Kirti 1820m)
39. \*B\* -  $31^{\circ}\text{C}$
40. \*515\* boxes ( $125360+93515=218875/425=515$ ) - Option mismatch, correct is 515
41. \*B\* - 1 hour 27 minutes
42. \*D\* - None of these (Actual total 115.19 kg)
43. \*C\* - 336 envelopes

44. \*A\* - 3817 (1470+948+1399)  
45. \*C\* - 450 cm (LCM of 50,75,90)

### ACHIEVERS SECTION (46-50)

46. \*C\* - P-(iii); Q-(i); R-(iv); S-(ii)  
47. \*A\* - (i) 15, (ii) 4, (iii) 0  
48. \*C\* - Statement-I true but Statement-II false  
49. \*D\* - T F T F  
50. (p) \*A\* - 66 cm, (q) \*A\* - 162 cm<sup>2</sup>

Want me to make this into a printable PDF answer key with explanations for the Maths section?